VERA — Vision-Enabled Reasoning Agent — System Architecture

v1.76.1 — LCARS-inspired task workspaces; body-selected hardware and tracking

Root-owned deployment controller (separate from VERA)

vera update + authenticated System GUI
/run/vera-updater/control.sock
peer-checked fixed operations

vera-updater.timer
vera-updater-check.service
local-time installation window

vera-updater.service
/opt/vera-updater/src/updater
stage / deploy / verify / recover

GitHub stable releases
mazum01/desktop-assistant
HTTPS, exact tag/commit

deployment.readiness RPC
loaded version/revision/PID
fresh telemetry + service/hardware health

/var/lib/vera-updater
protected backups + transaction state
post-reboot continuation

NF-A6x25 PWM Fan
PWM ← GPIO13 (kernel HW PWM, 25 kHz)
TACH → GPIO6 (2 ppr)

Hardware (Raspberry Pi 5)

Waveshare ESP32-S3-Touch-LCD-1.47
172×320 SPI TFT (touch disabled)
(BLE mouth/status display + BLE OTA)
C6 touch + legacy profiles supported

Display microSD (S3)
/videos/*.mjpg or *.mjpeg
baseline JPEG streams, no audio

Optional QMI8658 (identity-probed)
not specified on S3 hardware
roll/pitch/relative yaw, no compass

reSpeaker XVF3800 output →
attached speaker

Pan servo + optional tilt
PCA9685 configurable channels
180deg direct-drive / legacy DS3218
configured safe travel limits

Pi Cam Module 3 Wide
(CSI-0)

reSpeaker XVF3800
(USB mic array)

Hailo AI Accelerator
(Hailo-8 / Hailo-10H, PCIe)

TMP117
(VERA3, I²C 0x48)

TMP102
(VERA, I²C 0x48)

Drivers (src/)

motion/head_tracker.py (original body)
legacy Kalman + minimum-jerk pan tracking
new-body idle/person-seek controller

motion/visual_servo.py (new body)
stable target lock + damped visual feedback
identified camera/servo Jacobian
config/tracking_calibration.json (local)

core/body.py
config/body.yaml (machine-local identity/tuning)
config/body_profiles.yaml (hardware/controller)
body-tagged runtime servo geometry

motion/servo_controller.py

vision/camera.py
(picamera2)

audio/input.py

perception/hailo_models.py
config/hailo/models.yaml
probe / select / download HEFs
python3 -m ...hailo_models --install

audio/version_announcer.py

perception/hailo_inference.py
(VStreams + InferModel API)

thermal/tmp117.py

perception/face_detector.py
(SCRFD decode + Haar fallback)

audio/tts.py
(espeak-ng)

perception/object_detector.py
(YOLOv11l on-chip NMS; YOLO26m+ONNX H8 fallback)

thermal/temperature_sensor.py
(probes TMP117 / TMP102)

thermal/fan.py
(sysfs HW PWM
+ lgpio fallback)

thermal/fan_tach.py
(lgpio edge cb)

audio/output.py

thermal/thermal_manager.py

External processes

firmware/upload_esp32_display_ble.sh
+ nimbleota.py uploader
(BLE firmware push, no USB)

scripts/desktop-assistant
CLI: ping / status / pan / say / watch / ...
display video list/play/stop/status
waits for correlated device reply

3rd-party subscribers
(dashboards, future ML pipelines)

~/.local/share/desktop-assistant/
telemetry.db (SQLite ring)
faces.db (identity registry)
thumbs/ (64×64 JPEG face crops)

Web Browser
http://<pi>:8080

systemd: desktop-assistant-core.service

RoomService
← room.set
→ room.updated / av.say
persists room name · visual scene
baseline histogram · sustained
divergence detection → TTS prompt

Arduino firmware
firmware/vera_display
(ESP32-S3 process boundary; C6 profiles retained)
sd_video + JPEGDEC / orientation fusion
main-loop queue + reserved angle footer
NimBLEOta service (BLE OTA, 0x8018)

MonoDepthService
scdepthv3 on Hailo-8 / Hailo-10H
→ vision.mono_depth_map
relative 320×256 depth, 3 Hz

IPCBridge (core)
PUB ipc:///tmp/desktop-assistant.pub
REP ipc:///tmp/desktop-assistant.rep
SUB ← thermal.pub (upstream)
cmds: ping / topics / last / publish / status

TelemetryService
← thermal.* / motion.position / audio.level
→ telemetry.flush
SQLite ring buffer

TrackingService (body-selected controller)
original: legacy Kalman/minimum-jerk pan only
new: fresh-frame visual pan/tilt servo
← perception.faces (atomic frame time/geometry)
← motion.position / settings / limits_changed
→ motion.pan_to / tilt_to (new body only)
← tracking.start_autotune / cancel_autotune (new)
→ motion.calibration_lock / av.say
← av.spoke (announce before probes)
→ tracking.debug / autotune_progress / autotune_done
Tilt: travel limits and center +/-20deg
idle random pan · speaking nod

MotionService
shared PCA9685, independent pan / tilt
50Hz bounded quintic profiles
← motion.pan_to / tilt_to / center / relax / stop
← motion.calibration_lock (expiring lease)
← motion.set_channels / set_limits / set_tilt_limits
→ motion.position / moved / settings / error
→ motion.channels_changed / limits_changed / tilt_limits_changed
RPC motion.get_status / motion.set_channels

VisionService
→ vision.frame_ready
← vision.capture_still
+ latest_frame()

WebService (FastAPI)
GET / — dashboard SPA
GET /stream — MJPEG 640×360 feed
WS /ws — live JSON status
GET /api/faces — list (has_thumb flag)
GET /api/skills — registered skills
GET /api/faces/{id}/thumb — face JPEG
PUT /api/faces/{id} — rename
DEL /api/faces/{id} — delete one
DEL /api/faces — delete all
GET|PUT /api/settings/quiet-hours
GET|PUT /api/settings/audio
GET|PUT /api/settings/voice
GET|PUT /api/room
GET|PUT /api/music/* — proxied via IPCClient
GET|POST /api/podcasts/* — proxied via IPCClient
POST /api/say|pan|version|utterance
port 8080

AudioCaptureService
→ audio.level / chunk
+ latest_chunk()

PerceptionService
← vision.frame_ready
→ perception.faces / perception.error
+ FaceEmbedder (ArcFace) + FaceRegistry

VoiceCommandService
← audio stream/VAD
→ voice.state / wake / transcript / intent
→ av.utterance (routed transcript)
wake-window + dialog context

AVService
← av.say / beep / utterance / announce_version
→ av.spoke / av.speaking_started / version_announced

ObjectService
← vision.frame_ready / object.set_enabled
→ object.detections
2fps · max_objects cap · BGR→RGB fix

DisplayService
← system.startup_status / service.started / *.error
← display.spectrum / av.speaking_started / av.spoke
← display.video (list/play/stop/status)
→ display.status / display.error / display.reply
→ display.reply.<request_id> / display.device_event
BLE command writes + framed notifications

PlaybackSpectrumService
← music.state_changed / podcast.playback
pw-record on sink monitor + FFT
→ display.spectrum

FaceService
← perception.faces / face.meet
→ av.say (greetings)
→ face.identified
cooldown-based re-greet

ClockService
+ ClockAnnouncer
→ av.say (hourly/half-hourly)
respects QuietHours

SkillsService
← av.utterance / thermal.temp
→ av.say / vision.describe / av.tell_joke
→ face.meet / motion.pan_to / music.*
→ object.set_enabled / music.set_volume
→ tracking.set_face_tracking
SkillRegistry — first-match dispatch
17 skills: help · system_status · volume
quiet_hours · weather · reminder
smart_home · news + 9 others
ConfigField framework: enabled / config_schema

NotificationService
← thermal.temp / perception.faces
← av.speaking_started / av.spoke
→ av.say (thermal alerts, absence alerts)
rate-limited · quiet-hours aware

ThermalService
— thermal.temp
— thermal.fan
— thermal.rpm
— thermal.critical
— thermal.error

IPCBridge (media)
PUB ipc:///tmp/desktop-assistant-media.pub
REP ipc:///tmp/desktop-assistant-media.rep
RPCs: music.get_state/set_volume/...
podcast.search/subscribe/play/...

systemd: desktop-assistant-thermal.service

IPCBridge (thermal)
PUB ipc:///tmp/desktop-assistant-thermal.pub
REP ipc:///tmp/desktop-assistant-thermal.rep

MusicService
← music.play / stop / pause / next
← music.thumbs_up / thumbs_down / set_volume
Pandora / audio playback

PodcastService
Apple Podcasts search + RSS subscriptions
← podcast.play/pause/resume/stop/refresh
→ podcast.playback / subscriptions_updated

Core (src/core/)

core MessageBus
(in-process pub/sub)

version.py
(reads /VERSION)

Service base
(start/tick/stop)

QuietHours
(enabled / start / end
loaded from config/quiet_hours.yaml)

thermal MessageBus
(in-process pub/sub)

media MessageBus
(in-process pub/sub)